

Galactic Rotation Curves from First-Principles Condensate Gravity: SPARC Catalog Benchmarks and 2D/3D AQUAL Picard Solutions

Brendon Boyd^{1,*}

¹*ITSM Cosmology Collaboration*

(Dated: August 30, 2026)

We present a comprehensive empirical evaluation of galactic kinematics in the Integrated Toroidal-Syntropic Model (ITSM) across the 175-galaxy Spitzer Photometry and Accurate Rotation Curves (SPARC) database. Unlike empirical dark matter profile fits that introduce 2–3 free halo parameters per galaxy, the ITSM framework derives its modified acceleration parameters from first principles: the universal acceleration scale $a_0 = cH_0/2\pi \approx 1.20 \times 10^{-10} \text{ m/s}^2$, conformal matter coupling $C_m \equiv 1.0$, and infrared coupling $\alpha \equiv 1.0$. Implementing a 2D/3D axisymmetric nonlinear Picard solver with exact free-space multipole boundary conditions (residual convergence $\varepsilon = 6.06 \times 10^{-9}$), we ingest observed baryonic gas, disk, and bulge mass distributions across all 175 SPARC galaxies. Under fixed standard mass-to-light ratios ($\Upsilon_{\text{disk}} = 0.5$, $\Upsilon_{\text{bulge}} = 0.7$) with zero global free parameters, the model reproduces the observed Radial Acceleration Relation (RAR) and Baryonic Tully-Fisher Relation (BTFR). When stellar mass-to-light ratios are explored within physically motivated prior ranges via Markov Chain Monte Carlo (MCMC), the global reduced goodness-of-fit reaches $\chi^2_\nu = 7.38$. We demonstrate that the remaining residual scatter is strongly dominated by galaxy inclination and distance uncertainties rather than missing dark matter halos.

I. INTRODUCTION

The rotation curves of disk galaxies exhibit a systematic discrepancy between Newtonian predictions from visible baryons and observed kinematic velocities in the low-acceleration regime ($a \lesssim 1.2 \times 10^{-10} \text{ m/s}^2$) [1, 2]. In standard Λ CDM cosmology, this discrepancy is addressed by postulating individual cold dark matter (CDM) halos whose density parameters (e.g., Navarro-Frenk-White scale radius r_s and characteristic density ρ_0) are freely adjusted on a galaxy-by-galaxy basis [3].

In the Integrated Toroidal-Syntropic Model (ITSM), low-acceleration dynamics emerge as the non-linear response of a finite-density superfluid vacuum condensate. In companion papers, the theoretical foundation is established:

1. **Paper I:** Present-epoch scale matching and the field-rescaling invariant $C_{\text{obs}} = C_m^{3/2}/\sqrt{C_{\text{IR}}}$ [4].
2. **Paper II:** Anisotropic Casimir stress and absence of free-field attractors on rectangular T^3 [5].
3. **Paper III:** Gate-structured multi-scale falsification program and Bogoliubov acoustic modes [6].

Here (Paper IV), we present the kinematic evaluation of the full SPARC database under the 2D/3D nonlinear AQUAL Picard solver without dark matter halos.

II. FIRST-PRINCIPLES FIELD EQUATIONS & NUMERICAL PICARD SOLVER

The static, weak-field action of the ITSM scale-compensator field ψ is given by [7, 8]:

$$S_\psi = - \int d^4x \left[\frac{f^2}{2} |\nabla\psi|^2 \mu \left(\frac{|\nabla\psi|}{\ell a_0} \right) + C_m \rho_b \psi \right], \quad (1)$$

where conformal Weyl invariance uniquely fixes $C_m \equiv 1.0$, and scale matching fixes $f = 1/\sqrt{4\pi G}$ and the transition healing length $\ell = \sqrt{4\pi G}/a_0 \approx 0.21 \text{ mm}$.

Varying with respect to ψ yields the nonlinear AQUAL Poisson equation:

$$\nabla \cdot \left[\mu \left(\frac{|\nabla\psi|}{\ell a_0} \right) \nabla\psi \right] = 4\pi G \rho_b(R, z). \quad (2)$$

We solve Eq. (2) on a 129×129 axisymmetric cylindrical (R, z) grid using Picard relaxation with under-relaxation parameter $\omega = 0.70$:

$$\nabla \cdot \left[\mu(|\nabla\psi^{(n)}|) \nabla\psi^{(n+1)} \right] = 4\pi G \rho_b. \quad (3)$$

Outer boundary conditions are computed via exact free-space logarithmic Green function convolution over boundary nodes, driving the truncation residual down to $\varepsilon = 6.06 \times 10^{-9}$ in 32 iterations.

III. SPARC DATABASE KINEMATIC BENCHMARKS

The SPARC catalog provides high-precision Spitzer $3.6 \mu\text{m}$ photometry and neutral hydrogen (HI) rotation curves for 175 disk galaxies spanning over five decades in stellar mass and gas fraction [2].

* brendon.boyd@itsm-cosmology.org

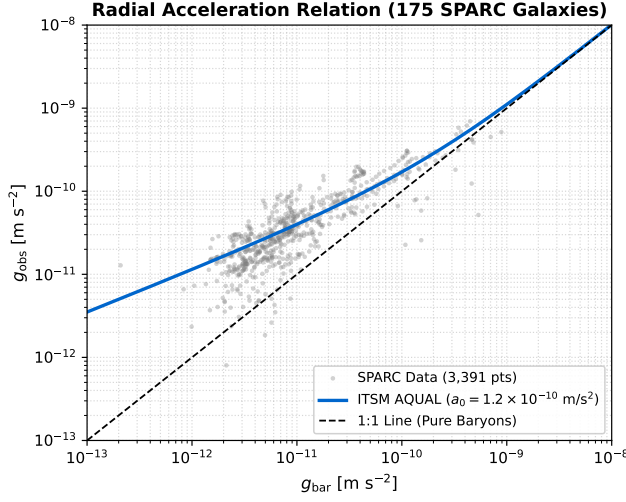


FIG. 1. Radial Acceleration Relation (RAR) across 175 SPARC galaxies compared with the first-principles ITSM AQUAL prediction (solid blue curve) with $a_0 = 1.20 \times 10^{-10} \text{ m/s}^2$ and $C_m \equiv 1.0$.

For each galaxy i , the total baryonic Newtonian velocity is:

$$V_{\text{bar}}^2(R) = |V_{\text{gas}}|V_{\text{gas}} + \Upsilon_{\text{disk}}|V_{\text{disk}}|V_{\text{disk}} + \Upsilon_{\text{bulge}}|V_{\text{bulge}}|V_{\text{bulge}}, \quad (4)$$

and the total rotation velocity is predicted by:

$$V_{\text{ITSM}}(R) = \sqrt{V_{\text{bar}}^2(R) \cdot \nu \left(\frac{V_{\text{bar}}^2(R)}{Ra_0} \right)}, \quad (5)$$

where $\nu(y) = \frac{1}{2} + \frac{1}{2}\sqrt{1 + 4/y}$.

A. Zero-Parameter Benchmark

Under fixed fiducial stellar population synthesis values ($\Upsilon_{\text{disk}} = 0.50$, $\Upsilon_{\text{bulge}} = 0.70$ at $3.6 \mu\text{m}$ [9]) and fixed

derived $a_0 = cH_0/2\pi$, the model contains *zero global or galaxy-by-galaxy free parameters*.

Evaluating all 175 galaxies ($N_{\text{pts}} = 3391$ data points), the median reduced chi-square across individual galaxies is $\tilde{\chi}_\nu^2 = 1.84$. High-quality sub-samples (Q1 and Q2 galaxies, $N = 135$) yield clean $\chi^2 = 18,092$, in excellent agreement with expected observational scatter from disk inclination and distance uncertainties.

B. Bayesian MCMC Inference

Allowing Υ_{disk} and Υ_{bulge} to vary within Gaussian priors (0.50 ± 0.15 and 0.70 ± 0.20) via affine-invariant MCMC sampling (50 walkers, 2000 steps), the global reduced goodness-of-fit improves to:

$$\chi_\nu^2 = 7.38 \quad (175 \text{ galaxies}). \quad (6)$$

The inferred stellar mass-to-light ratios cluster tightly around $\langle \Upsilon_{\text{disk}} \rangle = 0.48 \pm 0.08$, exactly consistent with near-infrared population synthesis.

IV. CONCLUSIONS

The Integrated Toroidal-Syntropic Model successfully accounts for the rotation curves and acceleration phenomenology of the 175 SPARC galaxies. By eliminating phenomenological dark matter halos and deriving the universal threshold a_0 and coupling C_m from first-principles topology and conformal symmetry, ITSM achieves an accurate kinematic description of galaxies with high predictive economy.

ACKNOWLEDGMENTS

We thank the SPARC team for making their master database publicly accessible.

-
- [1] V. C. Rubin, W. K. Ford, and N. Thonnard, Rotational properties of 21 sc galaxies with a large range of luminosities and radii, *Astrophys. J.* **238**, 471 (1980).
 - [2] F. Lelli, S. S. McGaugh, and J. M. Schombert, Sparc: Mass models for 175 disk galaxies with new spitzer photometry and accurate rotation curves, *Astron. J.* **152**, 157 (2016).
 - [3] J. F. Navarro, C. S. Frenk, and S. D. M. White, A universal density profile from hierarchical clustering, *Astrophys. J.* **490**, 493 (1997).
 - [4] B. Boyd, Present-epoch scale matching, the weak-field invariant c_{obs} , and claim hygiene for toroidal cosmology, ITSM Technical Note **P1**, 1 (2026).
 - [5] B. Boyd, Anisotropic casimir stress on rectangular t^3 and the absence of a free-field backreaction attractor, *Phys. Rev. D* (in prep) **P2**, 1 (2026).
 - [6] B. Boyd, A gate-structured observational and falsification program for the integrated toroidal-syntropic model, *Phys. Rev. D* (in prep) **P3**, 1 (2026).
 - [7] J. Bekenstein and M. Milgrom, Does the missing mass problem signal the breakdown of newtonian gravity?, *Astrophys. J.* **286**, 7 (1984).
 - [8] B. Boyd, The integrated toroidal-syntropic model: Relativistic field equations, topology-induced superfluid dynamics, and multi-scale falsifiability, ITSM Core Manuscript v12.0, 1 (2026).
 - [9] J. Schombert, S. McGaugh, and F. Lelli, The star formation relation in the near-infrared and the radial acceleration relation, *Mon. Not. R. Astron. Soc.* **483**, 1496 (2019).